

## CITS3002: Lab 3

### Exercise 1:

- a) Write a Python program that asks the user to enter two binary numbers, each consisting of 8 bits long. The program must check each input to ensure it contains only the characters 0 and 1 and that its length is exactly 8 bits; if either input is invalid, the program should print an error message and terminate. If both inputs are valid, the program should compute the Hamming distance between the two numbers. Finally, the program should print the computed Hamming distance.
- b) Modify the program from Exercise 1 so that instead of receiving two inputs, it asks the user to enter four valid codewords, each consisting of 8 binary digits. The program must verify that each input contains only 0s and 1s and has a length of 8 bits; if any input is invalid, the program should print an error message and terminate. After receiving the four valid codewords, the program should compute and print the Hamming distance of the code formed by these codewords.

### Exercise 2:

Write a Python program that receives an 8-bit binary data word from the user and generates the Hamming code for single-bit error correction. The program must first verify that the input contains exactly 8 bits and only the characters 0 and 1; otherwise, it should print an error message and terminate.

- a) The program should output the complete Hamming codeword, which consists of the original 8 data bits and the 4 computed parity bits.
- b) By modifying the program from part a, write a Python program that receives a 12-bit Hamming codeword as input and detects whether an error has occurred in the code. The output of the program should simply indicate whether an error has occurred.

### Exercise 3:

- a) Write a Python program that simulates the basic operation of a Stop-and-Wait communication protocol when the communication channel is free of noise. The program should include two functions: sender and receiver. The sender has four frames (A, B, C, and D) and sends one frame at a time to the receiver. When a frame is received, the receiver should display the received frame and send an acknowledgment (ACK) back to the sender. After the sender receives the ACK, it should proceed to send the next frame. The program should continue this process until all frames have been successfully delivered to the receiver and acknowledged.
- b) Modify the program from Exercise 3-a to simulate a Stop-and-Wait protocol with transmission errors. Assume that the communication channel fails with a probability of 20%, meaning that either a transmitted frame or its ACK may be lost. For each frame, determine how many retransmissions are needed before it is successfully delivered to the destination.
- c) Repeat exercise 3-b with the channel failure probability increased to 50%, and compare the retransmission counts in the two cases for each frame.